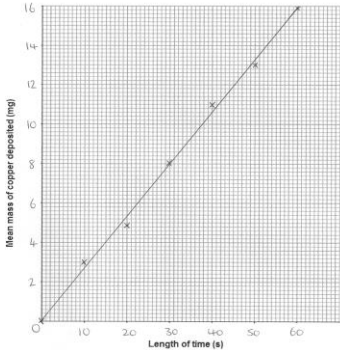


Common questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7/1	(a)			 award (2) for all points plotted correctly – tolerance $\pm\frac{1}{2}$ square award (1) for any four correct points award (1) for straight line through origin do not accept point to point line		3		3	3	3
	(b)	(i)		2.5 (2) ignore +/- if incorrect award (1) for correct workings e.g. $\frac{8.2-8.0}{8.0} \times 100 \text{ or } \frac{7.8-8.0}{8.0} \times 100 \text{ or } \frac{0.2}{8.0} \times 100$	1	1		2	2	2
		(ii)		some of the copper fell to the bottom of the beaker / some copper was left in the beaker time too long or too short – neutral some copper was left on the electrode – neutral reference to concentration of solution – neutral			1	1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	I	aluminium ions are positively charged and cathode is negative (1) opposites attract / move to opposite charge / gain three electrons from the cathode (1) aluminium 'atoms' – do not credit for first mark but second mark possible	2			2		
			II	$2 \text{ Al}_2\text{O}_3 \rightarrow \underline{\quad 4 \text{ Al} \quad} + \underline{\quad 3 \text{ O}_2 \quad}$ both products (1) correct balancing (1) balancing mark can only be awarded if both products are correct		2		2	1	
		(ii)		K_2CO_3		1		1		
				Question 7/1 total	3	7	1	11	6	6